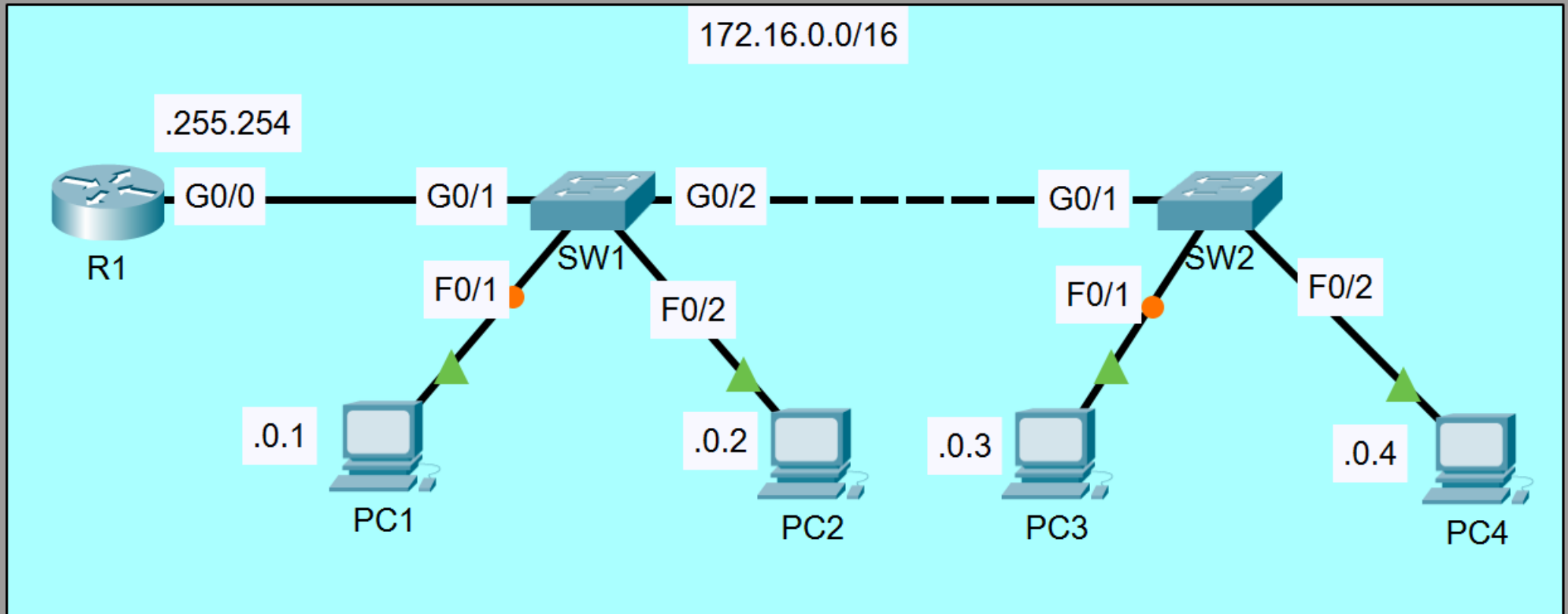


This is a single LAN connected to R1 g0/0 interface. The goal for today is to configure host names and IP addresses on the devices, as well as to disable unused interfaces on the switch





R1

Physical Config CLI Attributes

use. Delivery of Cisco cryptographic products does not imply third-party authority to import, export, distribute or use encryption. Importers, exporters, distributors and users are responsible for compliance with U.S. and local country laws. By using this product you agree to comply with applicable laws and regulations. If you are unable to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
<http://www.cisco.com/wwl/export/crypto/tool/stqrg.html>

If you require further assistance please contact us by sending email to export@cisco.com.

Cisco CISC02911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

Renaming Router to R1

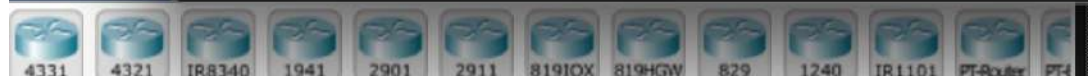
```
Router>Enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
```

Copy

Paste

Top

Root



Scenario 0

Fire Last Status Source Destination Type Col

SW2

Physical Config CLI Attributes

```
*      1      26      WS-C2960-24TT      12.2      C2960-LANBASE-M

Cisco IOS Software, C2960 Software (C2960-LANBASE-M), Version 12.2(25)FX, RELEASE SOFTWARE (fc1)
Copyright (c) 1986-2005 by Cisco Systems, Inc.
Compiled Wed 12-Oct-05 22:05 by pt_team

Press RETURN to get started!

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW2
SW2(config)#
```

Renaming Switch to SW2

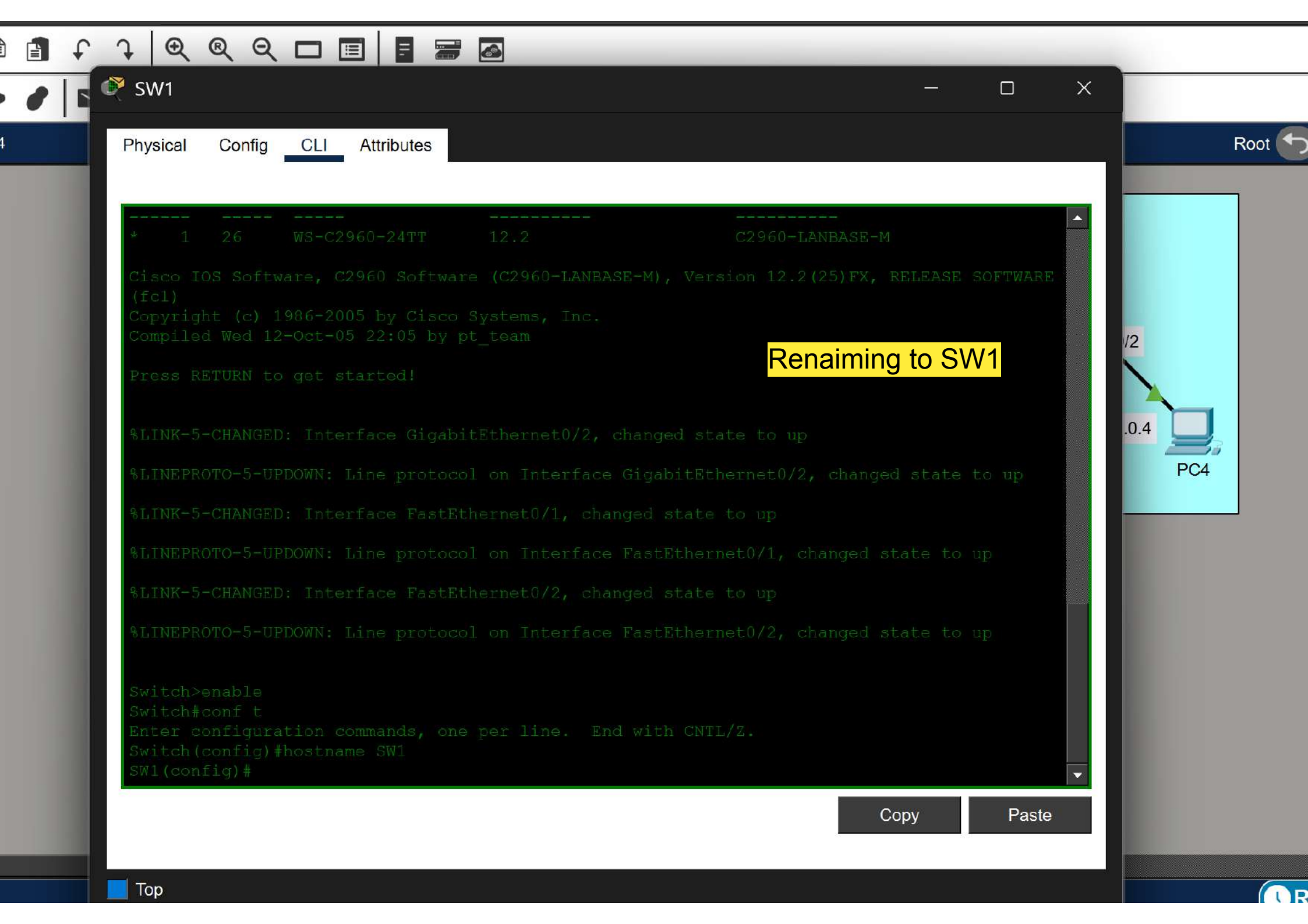
Copy Paste

Top

Root



Rea



SW1

Physical Config CLI Attributes

```
-----
*      1      26      WS-C2960-24TT      12.2      C2960-LANBASE-M

Cisco IOS Software, C2960 Software (C2960-LANBASE-M), Version 12.2(25)FX, RELEASE SOFTWARE
(fcl)
Copyright (c) 1986-2005 by Cisco Systems, Inc.
Compiled Wed 12-Oct-05 22:05 by pt_team

Press RETURN to get started!

%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#
```

Renaming to SW1

Copy

Paste

Top

Root

PC4

here I gave R1 G0/0 interface an IP address with a Class B or /16 prefix length

```
Router>Enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#do show ip int brief
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0       unassigned      YES unset  administratively down down
GigabitEthernet0/1       unassigned      YES unset  administratively down down
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down

R1(config)#int g0/0
R1(config-if)#ip address 172.16.255.254 255.255.0.0
R1(config-if)#
```

I then manually configured the speed and duplex of the interface , along with leaving a description of the interface.

```
GigabitEthernet0/1      unassigned      YES unset      administratively down down
GigabitEthernet0/2      unassigned      YES unset      administratively down down
Vlan1                   unassigned      YES unset      administratively down down

R1(config)#int g0/0
R1(config-if)#ip address 172.16.255.254 255.255.0.0

R1(config-if)#speed 1000
R1(config-if)#duplex full
R1(config-if)#desc
R1(config-if)#description ## TO SW1 ##
#
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

R1(config-if)#do show ip int brief
Interface                IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0       172.16.255.254  YES manual  up                    up
GigabitEthernet0/1       unassigned      YES unset   administratively down down
GigabitEthernet0/2       unassigned      YES unset   administratively down down
Vlan1                    unassigned      YES unset   administratively down down
R1(config-if)#
```



```
GigabitEthernet0/0      172.16.255.254  YES manual up      up
GigabitEthernet0/1      unassigned      YES unset  administratively down down
GigabitEthernet0/2      unassigned      YES unset  administratively down down
Vlan1                   unassigned      YES unset  administratively down down
```

```
R1(config-if)#int range g0/1-2
```

```
R1(config-if-range)#des
```

```
R1(config-if-range)#description ##Not In Use##
```

```
R1(config-if-range)#end
```

```
R1#
```

```
%SYS-5-CONFIG_I: Configured from console by console
```

```
write
```

```
Building configuration...
```

```
[OK]
```

```
R1#show startup-config
```

```
Using 785 bytes
```

```
!
```

```
version 15.1
```

```
interface GigabitEthernet0/0
```

```
description ## TO SW1 ##
```

```
ip address 172.16.255.254 255.255.0.0
```

```
duplex full
```

```
speed 1000
```

```
!
```

```
interface GigabitEthernet0/1
```

```
description ##Not In Use##
```

```
no ip address
```

```
duplex auto
```

```
speed auto
```

```
shutdown
```

```
!
```

```
interface GigabitEthernet0/2
```

```
description ##Not In Use##
```

```
no ip address
```

```
duplex auto
```

```
speed auto
```

```
shutdown
```

```
!
```

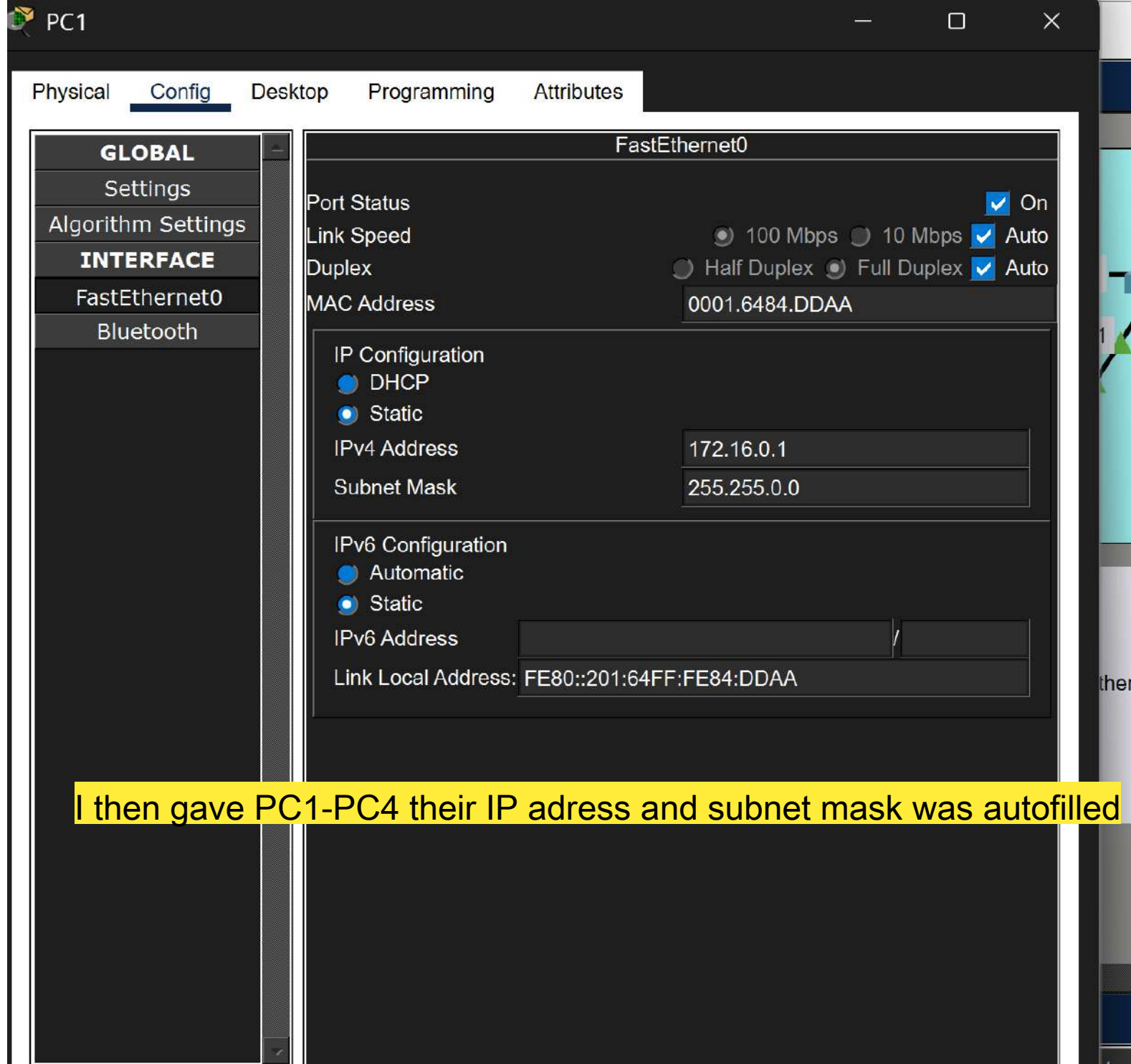
```
interface Vlan1
```

```
no ip address
```

```
shutdown
```

from PC1 to PC4 I configured the default gateway to the G0/0 interface of R1

Settings	Global Settings
Algorithm Settings	
INTERFACE	
FastEthernet0	Display Name PC1
Bluetooth	Interfaces FastEthernet0
	Gateway/DNS IPv4
	☐ DHCP
	☒ Static
	Default Gateway 172.16.255.254
	DNS Server
	Gateway/DNS IPv6
	☐ Automatic
	☒ Static
	Default Gateway
	DNS Server
	Device Clock: 10:18:03 Sun Feb 28 1993 UTC
	NTP
	Server Ip:
	Authentication
	☐ Enable ☒ Disable



Physical Config Desktop Programming Attributes**GLOBAL**

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ OnLink Speed ☐ 100 Mbps ☐ 10 Mbps ☒ AutoDuplex ☐ Half Duplex ☐ Full Duplex ☒ Auto

MAC Address 0030.F21C.46E3

IP Configuration

☐ DHCP☒ Static

IPv4 Address 172.16.0.2

Subnet Mask 255.255.0.0

IPv6 Configuration

☐ Automatic☒ Static

IPv6 Address

Link Local Address: FE80::230:F2FF:FE1C:46E3



PC3



Physical

Config

Desktop

Programming

Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status

☒ On

Link Speed

☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex

☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address

0003.E442.5E07

IP Configuration

☐ DHCP☒ Static

IPv4 Address

172.16.0.3

Subnet Mask

255.255.0.0

IPv6 Configuration

☒ Automatic☒ Static

IPv6 Address

Link Local Address: FE80::203:E4FF:FE42:5E07

here I followed a similar process to what I did with R1. I manually configured the speed and duplex of the interface connected with the router and SW2. however I did not manually configure the interfaces connected to the end hosts

10/100BaseTX

```
SW1(config)#int g0/1
SW1(config-if)#speed 1000
SW1(config-if)#duplex full
SW1(config-if)#desc
SW1(config-if)#description ##To R1##
SW1(config-if)#
SW1(config-if)#int g0/2
SW1(config-if)#duplex full
SW1(config-if)#speed 1000
SW1(config-if)#desc
SW1(config-if)#description ##to sw2##
SW1(config-if)#
```

```

10/100BaseTX
Fa0/19          notconnect    1          auto      auto
10/100BaseTX
Fa0/20          notconnect    1          auto      auto
10/100BaseTX
Fa0/21          notconnect    1          auto      auto
10/100BaseTX
Fa0/22          notconnect    1          auto      auto
10/100BaseTX
Fa0/23          notconnect    1          auto      auto
10/100BaseTX

```

```

SW1(config)#int g0/1
SW1(config-if)#speed 1000
SW1(config-if)#duplex full
SW1(config-if)#desc
SW1(config-if)#description ##To R1##
SW1(config-if)#
SW1(config-if)#int g0/2
SW1(config-if)#duplex full
SW1(config-if)#speed 1000
SW1(config-if)#desc
SW1(config-if)#description ##to sw2##
SW1(config-if)#
SW1(config-if)#
SW1(config-if)#
SW1(config-if)#int range f0/1-2
SW1(config-if-range)#desc
SW1(config-if-range)#description ##To End Host#
SW1(config-if-range)#int range f0/3-24
SW1(config-if-range)#desc
SW1(config-if-range)#description ##not in use##
SW1(config-if-range)#shutdown

```

disabling unused interfaces and leaving descriptions

```

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to administratively
down

```


this command was to double check the interface status, I then proceeded to save the configuration

```
SW1(config-if-range)#
SW1(config-if-range)#
SW1(config-if-range)#do sh int status
Port      Name           Status      Vlan      Duplex  Speed  Type
Fa0/1     ##To End Host# connected    1         a-full   a-100   10/100BaseTX
Fa0/2     ##To End Host# connected    1         a-full   a-100   10/100BaseTX
Fa0/3     ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/4     ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/5     ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/6     ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/7     ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/8     ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/9     ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/10    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/11    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/12    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/13    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/14    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/15    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/16    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/17    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/18    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/19    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/20    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/21    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/22    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/23    ##not in use## disabled     1         auto     auto    10/100BaseTX
Fa0/24    ##not in use## disabled     1         auto     auto    10/100BaseTX
Gig0/1    ##To R1##      connected    1         full     1000    10/100/1000BaseTX
Gig0/2    ##to sw2##     connected    1         full     1000    10/100/1000BaseTX
SW1(config-if-range)#
```

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same process with SW2, manually configured interface connected with SW1 and unused interfaces were shut down

```
SW2>enable
SW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#int g0/1
SW2(config-if)#speed1000
      ^
% Invalid input detected at '^' marker.

SW2(config-if)#speed 1000
SW2(config-if)#duplex full
SW2(config-if)#int f0/1-2
      ^
% Invalid input detected at '^' marker.

SW2(config-if)#int range f0/1-2
SW2(config-if-range)#desc ##To End User##
SW2(config-if-range)#int range f0/3-24
SW2(config-if-range)#desc
SW2(config-if-range)#description ##not in use##
SW2(config-if-range)#int g0/0
%Invalid interface type and number
SW2(config)#int g0/1
SW2(config-if)#desc
% Incomplete command.
SW2(config-if)#desc
SW2(config-if)#description
% Incomplete command.
SW2(config-if)#description ##to SW1##
SW2(config-if)#EIXT
      ^
% Invalid input detected at '^' marker.

SW2(config-if)#exit
SW2(config)#write
      ^
% Invalid input detected at '^' marker.

SW2(config)#exit
SW2#
%SYS-5-CONFIG_I: Configured from console by console
SW2#
```

Copy

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```
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to
administratively down
SW2(config-if-range)#
SW2(config-if-range)#end
SW2#
%SYS-5-CONFIG_I: Configured from console by console
```

```
SW2#
```

```
SW2#show int status
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Fa0/1	##To End User##	connected	1	a-full	a-100	
10/100BaseTX						
Fa0/2	##To End User##	connected	1	a-full	a-100	
10/100BaseTX						
Fa0/3	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/4	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/5	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/6	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/7	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/8	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/9	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/10	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/11	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/12	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/13	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/14	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/15	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/16	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/17	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/18	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/19	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/20	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/21	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/22	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/23	##not in use##	disabled	1	auto	auto	10/100BaseTX
Fa0/24	##not in use##	disabled	1	auto	auto	10/100BaseTX
Gig0/1	##to SW1##	connected	1	full	1000	
10/100/1000BaseTX						
Gig0/2		notconnect	1	auto	auto	
10/100/1000BaseTX						

```
SW2#
```